

1 Conductivity of electrolyte solutions

EXPERIMENT 1

Conductivity of electrolyte solutions

Goal

The goal of this experiment is to compare the conductivity of a set of pure substances, solutions, and mixtures of electrolytes. The overall goal is to understand how conductivity changes with the nature or reactive character of a substance and with its concentration.

Materials

- | | |
|--|--|
| ☐ jumbo test tubes | ☐ 0.1M solutions of KNO ₃ , KCl, KOH, HCl, and Ca(NO ₃) ₂ |
| ☐ conductivity meter | ☐ an unknown solution |
| ☐ 6M solution of NH ₃ | ☐ 5mL, 10mL, 25mL pipets (preferred Mohr's) |
| ☐ 6M solution of CH ₃ COOH | ☐ 250mL and 50mL beakers |
| ☐ Ethyl alcohol | ☐ rinsing water bottle |

Background

Quantitative studies of the conductivity of electrolyte solutions help us understand the nature of chemicals in solution. Conductivity measurements can help us obtain chemical properties such as solubility or PH. The conductive properties of a solution depend on factors such as the nature of the solvent or solute and temperature. It depends ultimately on ionic mobility and the capacity of ions to carry electric charge. This section addresses conductivity and how this property is measured in a chemistry laboratory.

Conductivity of pure substances

Conductivity is a type of electric conduction resulting from ion conduction, where ionic species, cations and anions, carry electrical charge after an electrical field is applied. Ionic conduction happens only when ions are free to move. Therefore, this type of electric conduction only happens either in molten ionic salts or in salt solutions. Solutions as metal conductors obey Ohm's Law and the resistance (R) of an ionic conductor can be computed using the same law, by applying a voltage (V) while measuring current intensity (I).

$$R = \frac{V}{I} \quad (1)$$

When dealing with liquid solutions it is convenient to speak in terms of conductance (G)—the inverse of resistance. The SI unit of resistance is the Ohm (Ω), whereas the corresponding unit for conductance is the Siemens (S), the inverse of the Ohm. In a metallic conductor, resistance depends on the conductor's length (l) and section (A). Resistance also depends on the characteristics of the metal in terms of its resistivity (ρ).

$$R = \rho \frac{l}{A} \quad (2)$$

In terms of conductance we have that this property depends on the inverse of the resistivity which is called the *conductivity* (κ)

$$G = \kappa \frac{A}{l} \quad (3)$$

The SI unit for κ is $\text{S} \cdot \text{m}^{-1}$ also written as $\Omega^{-1} \cdot \text{m}^{-1}$. As the conductivity of solutions is in general a small number, κ is regularly expressed as $\text{mS} \cdot \text{cm}^{-1}$ or $\mu\text{S} \cdot \text{cm}^{-1}$. For example, κ for distilled water ranges from 0.5 to 3 $\mu\text{S} \cdot \text{cm}^{-1}$, whereas that for tap water ranges from 50 to 800 $\mu\text{S} \cdot \text{cm}^{-1}$. This change in conductivity results from the existence of cations such as Ca^{2+} , Ca^{2+} , Mg^{2+} , K^{+} and ions such as HCO_3^{-} , Cl^{-} or SO_4^{2-} in tap water.

- 1 **Nonelectrolyte:** When the aqueous solution of that substance does not conduct electricity. In general terms, these electrolytes present low conductivities, lower than $10 \mu\text{S} \cdot \text{cm}^{-1}$.
- 2 **Weak electrolyte:** When the aqueous solution of that substance conducts electricity poorly. In general terms, these electrolytes present medium conductivities, between 10 - $1000 \mu\text{S} \cdot \text{cm}^{-1}$.
- 3 **Strong electrolyte:** When the conductivity of the aqueous solution of that substance is high. In general terms, these electrolytes present large conductivities, larger than $1000 \mu\text{S} \cdot \text{cm}^{-1}$.

Electrolyte conductivity

The conductivity of an electrolyte solution depends on the nature of the electrolyte and its concentration. Table salt and sugar are soluble in water. When sugar dissolves in water its molecules remain intact because water can not break apart the strong covalent bonds between their atoms. As a neutral molecule, sugar in solution will not conduct electricity and it is an example of a *nonelectrolyte*. The conductivity of nonelectrolyte solutions is very low. For example, at room temperature, the conductivity of ethanol $\text{CH}_3\text{CH}_2\text{OH}$, a nonelectrolyte is only $1 \mu\text{S} \cdot \text{cm}^{-1}$. Differently, table salt, sodium chloride or NaCl , is an ionic compound that gives place to two ions, Na^{+} and Cl^{-} when dissolved in water. Table salt is a *strong electrolyte*. The conductivity of strong electrolyte solutions is very large. For example, at room temperature, the conductivity of a 0.05M KCl solutions, a strong electrolyte just like table salt, is $5000 \mu\text{S} \cdot \text{cm}^{-1}$. These ions are responsible for the solution's conductivity. The number or more specifically the concentration, and the mobility of these ions will determine how conductive the solution is. Solutions of weak electrolytes such as NH_3 present a medium conductivity, higher than that for nonelectrolytes but lower than that for strong electrolytes. For example, at room temperature, the conductivity of a 0.06M acetic acid (CH_3COOH) solution is $1000 \mu\text{S} \cdot \text{cm}^{-1}$. Based on these observations we can classify the electrolytic nature of substances using conductivity measurements as shown in Figure 1.

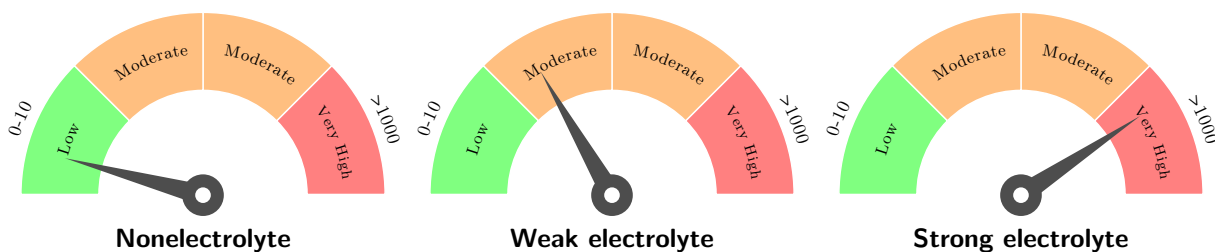


Figure 1 Three different conductivity measurements for three different electrolyte types. Low conductivities lower than $10 \mu\text{S} \cdot \text{cm}^{-1}$ correspond to nonelectrolytes, whereas very high conductivities, higher than $1000 \mu\text{S} \cdot \text{cm}^{-1}$ correspond to strong electrolytes.

Factors affecting conductivity

The conductivity of a pure substance depends on intrinsic and extrinsic factors. Intrinsic factors are determined by the nature of the substance which ultimately depends on the mobility of ions. Anything impeding this movement will create resistance to the current flow. Ionic movement is impacted by inter-ionic interaction, solute-solvent interactions, and solvent interactions. Temperature is an extrinsic factor that also plays a key role in ionic conductivity as high temperatures increase the average kinetic energy of ions hence favoring ionic conduction. The effect of concentration is easily appreciated in

isomolar solutions, which are solutions with the same molarity. The more ions in the solution, the higher the conductivity will be. For example, at room temperature, the conductivity of a 0.02M HCl solutions is $7\text{mS} \cdot \text{cm}^{-1}$, whereas the conductivity of a 0.1M HCl solutions is $44\text{mS} \cdot \text{cm}^{-1}$.

Additivity rule of conductivity

Interestingly, when two electrolyte solutions are mixed, the conductivity of the resulting solution will be equal to the sum of the two separate solutions with the same concentration as in the mixture. This is called the *additivity rule*. Conductance and conductivity are additive properties, which means that, at the same concentration, the conductivity of a solution resulting from adding two other solutions (κ_{mix}) corresponds to the sum of the conductivities of the two separate solutions.

$$\kappa_{\text{mix}} = \kappa_a + \kappa_b \quad (4)$$

This rule only holds in diluted conditions and assumes there is no interaction or chemical reaction between the species in the separate solutions. In particular, when two electrolytes undergo a chemical reaction, new products will form and the additivity rule will not apply. Hence, a large deviation from the sum of separate conductivities is an indication of a chemical reaction. The following example deals with the additivity rule for conductivities.

Example

The conductivity of a 0.050 M HNO_3 solution was 19.6 mS/cm , whereas that of a 0.050 M KCl solution was 9.6 mS/cm . A new solution is prepared by mixing 10 mL of 0.10 M HNO_3 and 10 mL of 0.10 M KCl . The conductivity of the resulting mixture was 29.2 mS/cm . Calculate the concentration of each species in the mix and determine whether the compounds have reacted.

Answer: HNO_3 and KCl are both strong electrolytes and hence they will dissociate producing: H^+ , NO_3^- , K^+ , and Cl^- . You can use the formula for dilutions to calculate the new concentration after mixing both reagents:

$$V_1 \cdot M_1 = V_2 \cdot M_2$$

where V_1 and M_1 are the initial volume and concentration and V_2 and M_2 are the values after the dilution. Therefore

$$M_2 = V_1 \cdot \frac{M_1}{V_2} = \frac{10 \text{ mL} \cdot 0.10 \text{ M}}{(10 \text{ mL} + 10 \text{ mL})} = 0.05 \text{ M}$$

Hence, we have that the concentration of all ions produced from the dissociation of the reagents is 0.05M. We can also see how the additivity rule works. We have that the conductivity (κ) of both separate reagents added is:

$$\kappa(\text{HNO}_3 \text{ 0.05 M}) + \kappa(\text{KCl 0.05 M}) = 19.6 \text{ mS/cm} + 9.6 \text{ mS/cm} = 29.2 \text{ mS/cm}$$

being this the same value as measured the conductivity of the mixture. As the additivity rule holds, there is hence no reaction between HNO_3 and KCl .

Ionic mobility

The conductivity of an electrolyte solution ultimately is related to ionic mobility. At the same time, ion *mobility* is related to their size. Small ions move fast which enhances conductivity, while large ions encounter more resistance to move and show lower conductivities. Table 2 displays a series of mobility values for positive and negative ions at infinite dilution and 298K.

Table 2 Ion mobilities u^∞ at infinity dilution* in $10^5 \text{ cm}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}$ *

Ion	u^∞	Ion	u^∞	Ion	u^∞	Ion	u^∞	Ion	u^∞
H_3O^+	363	Mg^{2+}	55	F^-	55	NO_3^-	74	ClO_3^-	65
Li^+	41	Sr^{2+}	60	Cl^-	80	CO_3^{2-}	75	CrO_4^{2-}	55
Na^+	52	Co^{2+}	53	Br^-	81	ClO_4^-	67	CHOO^-	54
K^+	76	OH^-	206	I^-	78	HCO_3^-	46	H_2PO_4^-	36

* Data listed at 25°C

Measuring conductivity

The conductivity of a solution is measured with a conductometer (see Figure 2). The principle behind a conductometer is simple. Conductivity is measured by measuring the resistance offered by the solution within two electrodes inside a conductivity cell. A Wheatstone bridge-type set up with alternating current is used to measure this resistance. The conductivity cell inside conductometers is made of a glass with two electrodes made of coated platinum. Conductivity cells are characterized by a cell constant K_{cell} that depends on the length and area of the cell.

$$K_{cell} = \frac{A}{l}$$



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Figure 2 A commercial conductometer.

Procedure

Part A. Getting started: standardize the conductimeter

- ☐ *Step 1:* – Install the conductivity meter. The probe of the instrument is immersed in a clean solution that must be kept aside until the end of the experiment. Mind the probe needs to be hydrated at all times.
- ☐ *Step 2:* – Look for the standard conductivity solution.
- ☐ *Step 3:* – To calibrate the conductivity meter, place 40mL of standard conductivity solution in a 50mL beaker while placing the probe in it.
- ☐ *Step 4:* – Make sure the conductivity cell is fully submerged in the liquid.
- ☐ *Step 5:* – Now press Setup (top right menu). Normal setups are: cell constant of 1.0, standard recognition with manual selected, reference temperature 25 degrees, temperature coefficient of 2.1%, and alarm limits off.
- ☐ *Step 6:* – Now press Standardize (top right menu) and input 1.000 mS/cm (or 1000 μ S/cm) in the keypad and press confirm (this should be the conductivity value of the standard). Finally, press Confirm (top right menu) to save the standard value. Now you are ready to measure.

Part A. Getting started: measuring conductivity

- ☐ *Step 1:* – It is very important to follow some cleaning steps before you measure the conductivity of any solution. You will need a 100 mL beaker labeled as waste, a wash bottle, and a 250 mL beaker with distilled water.

1. Rinse the probe using the wash bottle over the waste beaker.
2. Dip the probe 3-4 times into the 250 mL beaker with clean distilled water.
3. Dip the probe into the solution to be measured. Use a 50 mL beaker (or a jumbo test tube) with at least 25 mL of solution. Use the same volume for all conductivity measurements.
4. Repeat cleaning steps 1 and 2.

- ☐ *Step 2:* – It is critical to properly read the conductivity units from the conductimeter. Every solution would give a different conductivity value and sometimes a different unit. The conductivity units are written next to the measurement in the conductivity. It is critical you copy the measurement with the unit given by the tool.

Good Lab Practice

- ✎ To ensure the probe is well cleaned and the water in the 250 mL beaker remains clean, the conductivity meter should display a value close to that of the distilled water during the cleaning.
- ✎ If the conductivity value of the cleaning water differs from the one of distilled water, replace the water in the 250 mL beaker with fresh distilled water, and thoroughly clean the probe.

Part B. Pure substances.

- ☐ *Step 1:* – Following the procedure outlined above, measure the conductivity of distilled water. Record the value in the Results section.
- ☐ *Step 2:* – Following the procedure outlined above, measure the conductivity of tap water. Record the value in the Results section.
- ☐ *Step 3:* – Following the procedure outlined above, measure the conductivity of Ethyl alcohol ($\text{C}_2\text{H}_5\text{OH}$). Record the value in the Results section.
- ☐ *Step 4:* – Classify the substance as strong, weak, or nonelectrolyte.

Part C. Series of electrolytes.

- ☐ *Step 5:* – In a clean 50 mL beaker, add 20 mL of 0.10 M HCl and 20 mL of distilled water (this is a $\frac{1}{2}$ dilution). Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section.
- ☐ *Step 6:* – In a clean 50 mL beaker, add 20 mL of 0.10 M KOH and 20 mL of distilled water (this is a $\frac{1}{2}$ dilution). Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section.
- ☐ *Step 7:* – In a clean 50 mL beaker, add 20 mL of 0.10 M KCl and 20 mL of distilled water (this is a $\frac{1}{2}$ dilution). Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section.
- ☐ *Step 8:* – In a clean 50 mL beaker, add 20 mL of 0.10 M KNO_3 and 20 mL of distilled water (this is a $\frac{1}{2}$ dilution). Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section.
- ☐ *Step 9:* – In a clean 50 mL beaker, add 20 mL of 0.10 M $\text{Ca}(\text{NO}_3)_2$ and 20 mL of distilled water (this is a $\frac{1}{2}$ dilution). Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section.

- ☐ *Step 10:* – Find an unknown solution. In a clean 50 mL beaker, add 20 mL of the unknown solution and 20 mL of distilled water (this is a $\frac{1}{2}$ dilution). Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity.
- ☐ *Step 11:* – Classify all substances as strong, weak, or nonelectrolytes and identify the unknown compound based on the conductivity values. Know that the unknown can also be water.

Part D. Series of Dilutions.

- ☐ *Step 12:* – Measure the conductivity of a 0.1 M HCl solution (the one you used to prepare for the previous step). Write down the result in the Results section.
- ☐ *Step 13:* – Now you are going to prepare two dilutions. Starting from the concentrated 0.1 M solution of HCl, you are going to prepare 40mL volumes of the following dilutions: 0.050 M and 0.020 M. The first is a $\frac{1}{2}$ dilution, that is 20ml of acid needs to be mixed with 20ml of water to get 40mL of 0.050M solution, whereas the second is a $\frac{1}{5}$ dilution, that is 8ml of acid needs to be mixed with 32ml of water. Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers.
- ☐ *Step 14:* – Measure the conductivity of the two diluted HCl solutions (0.050 M, and 0.020 M) and record the values.

Part E. Mixtures of HCl and KNO₃.

- ☐ *Step 15:* – Write down the conductivity of the 0.050 M solution of HCl from the previous section in the Results section.
- ☐ *Step 16:* – Write down the conductivity of the solution of KNO₃ from the previous section in the Results section.
- ☐ *Step 17:* – In a clean 50 mL beaker, add 20 mL of 0.10 M HCl and 20 mL of 0.10 M KNO₃. Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section, minding that both solutions have now the same concentration and therefore only one number needs to be listed.

Part F. Mixtures of NH₃ and CH₃COOH.

- ☐ *Step 18:* – Prepare the following dilutions: 100mL of acetic acid (CH₃COOH) 0.12 M from 6M (add 2mL of concentrated solution and fill up with distilled water until 100mL) and 40mL of ammonia (NH₃) 0.12 M from 6M (add 2mL of concentrated solution and fill up with distilled water until 100mL). Carefully read the concentration of the initial concentrated solutions indicated in the labels. Work in the hood when handling highly concentrated solutions.
- ☐ *Step 19:* – In a clean 50 mL beaker, add 20 mL of the 0.12 M NH₃ dilution and 20 mL of distilled water (this is a $\frac{1}{2}$ dilution). Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section. Work in the hood when handling highly concentrated solutions.
- ☐ *Step 20:* – In a clean 50 mL beaker, add 20 mL of 0.12 M CH₃COOH and 20 mL of distilled water (this is a $\frac{1}{2}$ dilution). Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section.
- ☐ *Step 21:* – In a clean 50 mL beaker, add 20 mL of 0.12 M NH₃ and 20 mL of 0.12 M CH₃COOH. Use a Mohrs pipet to transfer the liquids, making sure you rinse with distilled water between transfers. Mix well and measure the conductivity. Recalculate the concentration after the dilution and write it down in the Results section, minding that both solutions have now the same concentration and therefore only one number needs to be listed.

⚠ CAUTION!

- ⚠ Handle concentrated acid or bases with care to avoid chemical burns.
- ⚠ Are you wearing your goggles? Do you know where the eye-washer is?



STUDENT INFO

Name: _____ Date: _____

Pre-lab Questions

Conductivity of electrolyte solutions

1. Starting with a 6 M solution of NaCl, calculate the volume of solution and the volume of water necessary to prepare 10 mL of a 3 M solution of NaCl.
2. How can you dilute a 3 M solution to form a 1.5 M solution? (e.g. I would pick up X mL of the solution and YmL of distilled water.)
3. How can you prepare 40mL of acetic acid ($\text{HC}_2\text{H}_3\text{O}_2$) 0.12 M from 6M.
4. How can you prepare 40mL of ammonia (NH_3) 0.12 M from 6M.
5. Given the data below, classify the following electrolytes as strong, weak, or nonelectrolytes:

Electrolyte	Concentration	Conductivity	Type of electrolyte
CuSO_4	0.05M	$4000 \mu\text{S} \cdot \text{cm}^{-1}$	_____
FeCl_3	0.05M	$10000 \mu\text{S} \cdot \text{cm}^{-1}$	_____
$\text{C}_6\text{H}_{12}\text{O}_6$	0.05M	$9 \mu\text{S} \cdot \text{cm}^{-1}$	_____
CH_3COOH	0.05M	$320 \mu\text{S} \cdot \text{cm}^{-1}$	_____

STUDENT INFO

Name: _____ Date: _____

Results
EXPERIMENT 1**Conductivity of electrolyte solutions****Part B: Pure substances**

	Conductivity Units	Electrolyte type	
Distilled water	_____	_____	_____
Tap water	_____	_____	_____
C ₂ H ₅ OH	_____	_____	_____

Part C: Electrolytes

	Concentration (M)	Conductivity Units	Electrolyte type	
HCl	_____	_____	_____	_____
KOH	_____	_____	_____	_____
KCl	_____	_____	_____	_____
KNO ₃	_____	_____	_____	_____
Ca(NO ₃) ₂	_____	_____	_____	_____
Unknown		_____	_____	
Unknown # _____		Unknown identity _____		

Part D: Concentration effect			
Concentration (M) Conductivity Units			
HCl	_____	_____	_____
HCl	_____	_____	_____
HCl	_____	_____	_____
Part E: HCl+KNO ₃ mixture			
Concentration (M) Conductivity Units			
HCl	_____	_____	_____
KNO ₃	_____	_____	_____
HCl+KNO ₃	_____	_____	_____
Part F: CH ₃ COOH+NH ₃ mixture			
Concentration (M) Conductivity Units			
CH ₃ COOH	_____	_____	_____
NH ₃	_____	_____	_____
CH ₃ COOH+NH ₃	_____	_____	_____

STUDENT INFO

Name: _____ Date: _____

Post-lab Questions

Conductivity of electrolyte solutions

1. Based on your results on Part B, which conductivity was higher, the one for distilled water or for tap water? Why?
2. Based on your results on Part C, specifically the conductivity values of KCl and KOH, how does ion mobility affect conductivity (see table in the Background section)? Use your data to illustrate your answer.
3. Based on your results on Part D, how does concentration affect conductivity? Use your data to illustrate your answer.
4. Based on your results in Part E, determine if their compounds have reacted or not. Does the additivity rule hold for this case?
5. Based on your results in Part F, determine if their compounds have reacted or not. Does the additivity rule hold for this case?

